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Magnets Everywhere (Science — Physical: forces, non-contact forces)

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Worksheet

Part A

1. C (friction), N (gravity), C (hand pushing door), N (magnet pulling paper clip).
2. *The **magnet's** north pole was painted red, but the **compass's** needle **didn't** move.* (Two possessive apostrophes — one magnet, one compass — and one contraction. Accept *compass'* if taught that style.)
3. e.g. *Although it was not touching the magnet, the paper clip moved. / The paper clip moved because the magnet pulled it without touching it.* (Look for a correct subordinating conjunction and, if the clause is fronted, a comma.)

Part C

1. Because it can push or pull without touching the object. Example: a fridge magnet holds a sheet of paper firmly in place (the force works through the paper); or it works through plastic and water.
2. ✓ iron nail · × aluminium foil · ✓ steel spoon (most steel; some stainless steel is not magnetic — accept either with a reason) · × copper wire · ✓ nickel · × glass marble · × wooden peg
3. Magnets only attract some metals — iron, nickel, cobalt and most steel. Other metals such as aluminium and copper are not attracted, e.g. a magnet does nothing to most drink cans because they are aluminium.
4. B **repel** (S faces S) — arrows pointing away from each other. C **repel** (N faces N) — arrows apart. D **attract** (N faces S) — arrows towards each other.
5. *Like poles (north and north, or south and south) repel each other.*
6. The needle should be drawn along the dashed field line at X, red end pointing towards the North Pole (up-left, roughly towards the top of the Earth). Explanation: the needle is a small magnet that lines up with Earth's magnetic field, so its north end points roughly north wherever it is held flat and still.
7. On a cloudy night sailors cannot see the stars (or the Sun) to find direction, but a compass works in any weather because it uses Earth's magnetic field, not the sky.
8. The crane driver can switch it off to drop the load exactly where they want; an ordinary magnet would stay stuck to the metal and the load would be hard to pull off.
9. Scientists are not completely sure yet — "researchers are still working out exactly how". The writer uses cautious words (*think, may*) so they don't state something as fact when it isn't fully proven (links to Week 2 modal verbs and fact vs opinion).

Connections

10. Horseshoe (12), bar (8), wand (6), ring (4), fridge (2).
11. a) 12 − 2 = **10** more b) 12 ÷ 4 = **3 times** c) 8 + 12 + 4 + 6 + 2 = **32** clips
12. Any two: same size/type of paper clips; hang clips from the same place (a pole/end of each magnet); same way of adding clips (one at a time, gently); same stand/height; nothing else magnetic nearby; same person counting. The only thing changed is the magnet.
13. Column graph: title (e.g. *Paper clips held by different magnets*); horizontal axis = magnet type; vertical axis = number of paper clips, scale in 1s or 2s from 0 to at least 12; five columns of equal width with gaps; heights 8, 12, 4, 6, 2.
14. a) Steel 9/12 = **3/4**; aluminium 3/12 = **1/4** b) 1/4 of 120 = **30** aluminium cans
15. **Bin A.** The magnetic roller **attracts** the steel cans, so they cling to it as it turns and drop off underneath, into Bin A. Aluminium is not attracted, so aluminium cans fly off the end of the belt into Bin B.
16. **Lookout**, grid reference **F4** (B1 → 3 north → B4 → face east → 4 squares → F4).
17. e.g. *From the jetty, walk 4 squares east along the bottom row to F1. Then turn north and walk 5 squares up the right-hand edge to the Lighthouse (F6).* (North first also works: 5 north to B6, then 4 east.) Look for correct compass directions and at least one adverbial of place.
18. Maglev trains don't touch the track — magnets lift them above it — so there are no wheels rubbing on rails and very little friction. With less friction slowing it down, the train can travel faster (and more smoothly).

Part D

19. a) how b) when (accept *how long*) c) where d) how e) when f) why
20. Any two: *gently*, *neatly*, *in curved lines* (¶3); *freely*, *slowly*, *flat and still*, *roughly* (¶4). (*roughly* tells "how exactly"; accept.)
21. Any two: *Surprisingly, · Deep inside, · For hundreds of years, · Inside the rubber seal of most fridge doors, · At scrapyards, · In recycling plants, · Instead, · Since the early 2000s,*
22. a) *During the storm, the compass needle swung round wildly.* (or *Wildly, the compass needle swung round during the storm.*) b) *Near a magnet, iron filings line up in curved lines.* (or *In curved lines, …*)
23. It warns the reader that the next fact is unexpected — most people think magnets stick to all metals — so it makes them pay attention / challenges a common belief.
24. a) firmly b) gradually c) instantly d) gently e) rapidly

Part E

25. a-3, b-1, c-5, d-2, e-4, f-6 (7 is a distractor — it describes a *contact* force).
26. e.g. *When I pushed the two north poles together, they repelled each other strongly.*

Part F — marking notes

Model (Option 1, ~120 words):
A compass is a simple tool that helps people find their way. Inside it, a thin magnetic needle can spin freely. Deep inside, Earth behaves like a giant magnet, so when you hold the compass flat and still, the needle slowly turns until one end points roughly north. Because the needle always settles in the same direction, you can easily work out north, south, east and west. For hundreds of years, sailors have relied on compasses, especially on cloudy nights when they could not see the stars. Today,

bushwalkers still carry them because they need no batteries. As a result, a compass can guide you safely home even when other equipment stops working.

Model (Option 2, ~110 words): *In recycling plants, steel and aluminium cans arrive completely mixed together. Before they can be recycled, they must be separated. First, the cans travel steadily along a conveyor belt past a strong magnet. Because steel is made mostly of iron, the magnet attracts the steel cans and pulls them quickly out of the stream. Meanwhile, the aluminium cans keep moving along the belt, because aluminium is not magnetic. Finally, each metal drops into its own bin. As a result, both metals can be melted down separately and made into new products. Without magnets, workers would have to sort millions of cans slowly by hand.*

A strong response: opening statement; logical how/why sequence with cause-and-effect connectives; accurate science vocabulary; ≥4 underlined adverbials (incl. a fronted one with a comma); a precise adverb (*steadily, freely, safely*); a complex sentence; a concluding sentence about why it matters. Watch for: commas missing after fronted adverbials; the misconception that magnets attract all metal; saying a compass points "to the North Pole exactly".

Part G

27. a) Nothing **else** pulled the steel cans — only the magnet did it (*only* describes the magnet). b) The magnet pulled the steel cans and **nothing else** — it didn't pull the aluminium cans (*only* describes the cans). Moving *only* changes which word it limits.
28. e.g. *First, he holds the north end of magnet 1 near one end of magnet 2. If they repel, that end of magnet 2 is also north, because like poles repel. If they attract, that end is south, so the north pole is the other end. After that, he checks the other end to make sure it does the opposite. Finally, he labels the north pole.* (Note: attraction alone is a weaker test — a magnet will also attract plain steel — so repulsion is the reliable sign. Strong students may mention this.)
29. Yes — magnets are found in many places: fridges (¶1/¶5), compasses and Earth itself (¶4), speakers, scrapyards and recycling plants (¶5), trains (¶6) and even inside animals' senses (¶7). Accept a reasoned "partly" (e.g. maglev trains are only in a few places).

Fast-finisher bank

F1. How: *carefully, silently* · When: *every morning* (accept how often), *at midnight* · Where: *at the scrapyard, under the table, beside the compass* · Why: *because it was cloudy, so that the door stays shut* · How often: *twice a week, every morning*. (*Every morning* can sit in When or How often — accept with reason.)
F2. e.g. a) *The magnet picked up the pins effortlessly.* b) *The turtles swam across the Indian Ocean.* c) *Early in the morning, the scrapyard crane lifted the old car.*
F3. a) A natural magnet — a dark rock (a kind of magnetite) that pulls on iron. b) Any three: *long ago* (when), *from a thread* (where), *on water* (where), *slowly* (how), *every time* (how often/when), *later* (when), *in China* (where), *about a thousand years ago* (when), *over time* (when), *today* (when), *especially when other equipment stops working* (when). c) Hanging or floating it meant there was very little friction, so the weak turning force from Earth's magnetic field could swing it round; on a table, friction would stop it turning. d) Historians aren't completely certain of the exact date or details, so the writer is careful not to state it as a definite fact.
F4. Change: number of cardboard sheets (e.g. 0, 2, 4, 6, 8). Measure: whether the paper clip is still pulled / the greatest distance at which the magnet pulls the clip / how many clips it holds through the card. Keep the same: same magnet, same paper clip, same type/thickness of cardboard, same distance/starting position, same way of testing, repeat each test. Prediction e.g. *I predict the pull will get weaker as I add sheets, because the magnet is further from the clip.* (Teacher note: cardboard itself does not block magnetism — the force weakens mainly because the sheets increase the **distance**. Great discussion point.) Steps e.g. *1. Carefully place the paper clip on the table. 2. Slowly lower the magnet on top of two sheets of card above the clip. 3. Gently lift the card and record whether the clip comes too.*
F5. e.g. a) *Without warning, the powerful horseshoe magnet jerked suddenly towards the steel fence.* b) *Since the early 2000s, the sleek white maglev train has raced smoothly towards Shanghai's airport.* c) *During the storm, the battered brass compass reliably showed the frightened sailors the way.*
F6. Look for heading; three imperative instructions with adverbs (*Gently slide..., Slowly bring..., Carefully return...*); one accurate fact; a safety rule with reason (e.g. *Keep magnets away from phones and bank cards so that they are not damaged*; *Never put small magnets in your mouth, because swallowing them is very dangerous*).
F7. Check: first person; five adverbials (two fronted, with commas); personification; two accurate facts (e.g. compass: lines up with Earth's magnetic field, sailors used compasses for hundreds of years; fridge magnet: works through paper, attracted to steel; maglev: lifted by magnets, little friction). Watch for invented "facts" such as exact speeds.
F8. Hold the **end** of bar 1 against the **middle** of bar 2. A magnet's force is weakest in the middle and strongest at the poles. If bar 1 is the magnet, its pole will attract the middle of the plain steel bar strongly. If bar 1 is the plain steel, it will hardly be pulled by the middle of the magnet. Swap and compare: the bar whose **end** pulls strongly on the other's **middle** is the magnet.
F9. For: e.g. less friction means fast, smooth, quiet travel; could link the airport and city quickly; electric so no exhaust fumes from the train itself. Against: e.g. very expensive to build; needs a special new track; Perth already has a train network; few places use maglev. Rhetorical question e.g. *What if you could travel from the airport to the city in just a few minutes?* (Avoid made-up statistics.)

Homework

1. By watching the Sun by day and the stars at night. On cloudy days or stormy nights they could not see them, so ships could drift off course.
2. Big pieces of iron or steel can pull (attract) the needle and make it point the wrong way.
3. It needs no batteries, and it keeps working when electronic equipment fails.
4. Because a compass doesn't point *exactly* to the North Pole (Earth's magnetic north is not in quite the same place, and nearby iron can affect it) — the writer is being accurate and careful.
5. a) *carefully* — how b) *On cloudy days* — when c) *near the ship's wheel* — where
6. *After the storm, the needle slowly settled.*
7. e.g. *The sailor's compass was old, but it still pointed north.*
8. Typical results: ✓ fridge door, steel cutlery (most), tin-can lids (food tins), steel bottle caps, many screws/nails, steel pot (some); × aluminium foil, drink cans, wooden spoons, plastic containers, glass, copper-coloured coins, many stainless-steel sinks (often not magnetic). Accept any honest data.
9. Common surprises: stainless-steel items that don't stick; a steel food tin that does stick; a painted metal object that sticks. Look for *attract* and a material name.
10. Any genuine example.

Teacher notes

- **Common misconceptions:** "magnets attract all metals"; "bigger magnets are always stronger" (Mia's data deliberately invites discussion — shape and material matter); "the compass points to the North Pole exactly"; "cardboard/paper blocks magnetism" (it is mostly distance); "magnetism and gravity are the same thing" (both are non-contact forces, but different).
- **Pole puzzle subtlety (Earth):** the end of a compass needle that points north is called its *north-seeking* or north pole. Because unlike poles attract, Earth's magnetic pole in the Arctic actually behaves like a magnet's *south* pole. That is why the giant-magnet bar in the diagram is deliberately **unlabelled**. Only raise this with a student who asks.
- **Diagram for Q6:** the field-line arrows point northward along the surface (the horizontal direction a compass needle aligns with). The diagram is simplified (real field lines also dip into the ground at an angle).
- **Safety:** small strong (neodymium) magnets are dangerous if swallowed; keep magnets away from bank cards and electronics. Homework tells students not to test electronics.
- **If the student finishes early:** F4 can become a real investigation; F8 and Part G Q28 are excellent "think aloud" puzzles.
- **If the student struggles:** act out adverbials — say a verb (*walk*) and have them add *how* (*slowly*), *where* (*to the door*), *when* (*after lunch*). Use real magnets for Q4.
- **Sources / fact-check:**
  - Magnetic force is a non-contact force acting through non-magnetic materials (paper, plastic, water); attracted materials iron, nickel, cobalt and most steels; not aluminium, copper, wood, plastic, glass. *Note:* some stainless steels (austenitic, e.g. many sinks/cutlery) are not magnetic, hence "most kinds of steel". — standard school physics (e.g. Australian Curriculum Year 4 Physical sciences; Science Learning Hub NZ "Magnets").

- Like poles repel, unlike poles attract; force strongest at the poles; iron filings show field pattern. — standard.
- Earth's magnetic field (generated in the molten outer core) makes Earth behave like a giant bar magnet; compass needle aligns with it and points roughly north (magnetic north ≠ true north; declination in Perth is small, about 0–2°). Phrased as "roughly north" and "when you hold a compass flat and still" because compasses are unreliable near the magnetic poles. — Geoscience Australia (geomagnetism); British Geological Survey.
- Compasses used for navigation "for hundreds of years"; F3: Chinese use for ship navigation about a thousand years ago (Song dynasty texts, early 1100s; Shen Kuo described a magnetic needle in 1088). Phrased with "Historians think". Lodestone = naturally magnetised magnetite.
- Fridge door seals contain a flexible magnetic strip (most modern fridges); loudspeakers/headphones use permanent magnets with a coil; scrapyard lifting electromagnets are switched off to release loads; recycling facilities use magnets (overband or drum/roller magnets) to remove steel; aluminium is separated in other ways (e.g. eddy-current separators — not mentioned to students). Most Australian drink cans are aluminium; food tins are mostly tin-plated steel.
- Maglev: magnetic levitation lifts the train off the guideway and linear motors propel it — no wheel–rail friction. Shanghai Maglev (Transrapid technology) links Pudong International Airport with Longyang Road station; inaugurated 2002, regular commercial service from 2004 — phrased as "since the early 2000s". Described only as "one of the fastest passenger trains in the world"; no speeds given. ("Between Pudong Airport and the city" — strictly it ends on the edge of the city centre at Longyang Road; accurate enough at Year 4.)
- Animal magnetoreception: evidence for loggerhead and other sea turtles (e.g. Lohmann lab research) and migratory birds (e.g. European robins); mechanisms still under research — hedged with "scientists think", "may", "still working out". Green and loggerhead turtles nest along the Ningaloo coast (WA DBCA / Ningaloo Turtle Program).
- Homework: binnacle = the stand/housing for a ship's compass, near the helm; liquid-filled compasses damp needle movement; iron/steel deflects compasses (ships' compasses are corrected for this).
- Connections data (Mia's magnets) are invented sample data for practice, not real measurements; all arithmetic checked (difference 10, ratio 3, total 32; 9/12 = 3/4, 1/4 × 120 = 30; grid route B1 → B4 → F4).

- **Next week builds on:** adverbials of time and manner feed straight into procedural writing (*From Seed to Flour* — recipes and measurement vocabulary).